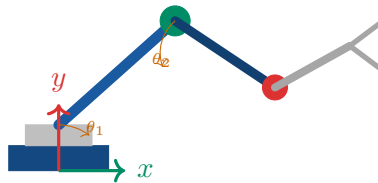


Introduction to Robotics & Intelligent Control Systems

Comprehensive End-Semester Study Notes

2nd Year Elective · All 7 Units Covered



“Robotics is the intersection of mathematics, mechanics, and intelligence.”

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Unit 1: Geometry & Coordinate Transformations

1.1 Fixed and Mobile Frames

Key Concept: Reference Frames

In robotics, we use two types of coordinate frames:

- **Fixed (World/Inertial) Frame $\{W\}$:** Stationary reference, denoted $\{0\}$.
- **Body/Mobile Frame $\{B\}$:** Attached to a moving body (link, end-effector), changes with motion.

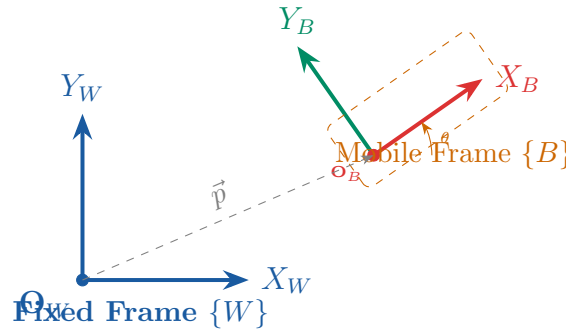


Figure 1: Fixed World Frame $\{W\}$ and Mobile Body Frame $\{B\}$. The vector $\vec{p}$ locates the origin of $\{B\}$ in $\{W\}$.

1.2 Coordinate Representation of Vectors

A vector $\vec{p}$ expressed in frame $\{A\}$:

$${}^A\vec{p} = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix}_A$$

The **same physical vector** has different numerical components in different frames. The coordinate transformation problem asks: *given ${}^B\vec{p}$, find ${}^A\vec{p}$.*

1.3 Rotation Matrices

Definition: Rotation Matrix

A rotation matrix $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ satisfies:

$$\mathbf{R}^{-1} = \mathbf{R}^T, \quad \det(\mathbf{R}) = +1$$

It represents the orientation of one frame relative to another. The set of all such matrices forms the **Special Orthogonal Group $SO(3)$** .

1.3.1 Rotation About the X-Axis (Roll)

$$\mathbf{R}_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}$$

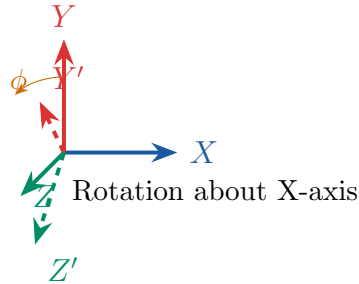


Figure 2: Rotation by angle ϕ about the X -axis. X -axis remains fixed; Y and Z rotate.

1.3.2 Rotation About the Y-Axis (Pitch)

$$\mathbf{R}_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

1.3.3 Rotation About the Z-Axis (Yaw)

$$\mathbf{R}_z(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

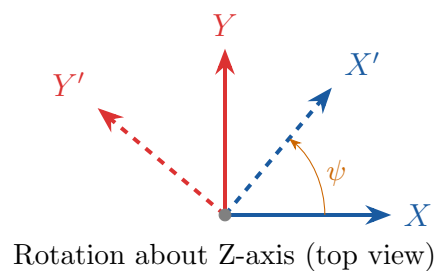


Figure 3: Rotation by ψ about Z -axis (viewed from above). The Z -axis points out of the page.

1.4 Composite Rotations

Important Property: Non-Commutativity

Rotations in 3D are **NOT commutative**:

$$\mathbf{R}_z(\psi) \mathbf{R}_y(\theta) \neq \mathbf{R}_y(\theta) \mathbf{R}_z(\psi)$$

The **order** of rotations matters! Always multiply from left to right for sequential rotations applied to the *fixed* frame, or right to left for the *body* frame.

For a sequence of rotations applied about the **current (body) frame** axes:

$$\mathbf{R}_{\text{total}} = \mathbf{R}_1 \cdot \mathbf{R}_2 \cdot \mathbf{R}_3 \quad (\text{post-multiply for body frame})$$

1.5 Yaw-Pitch-Roll (YPR) Representation

YPR (Euler Angles — ZYX Convention)

Yaw (ψ) $\rightarrow$ **Pitch** (θ) $\rightarrow$ **Roll** (ϕ), applied about *fixed axes* in Z-Y-X order:

$$\mathbf{R}_{ZYX}(\psi, \theta, \phi) = \mathbf{R}_z(\psi) \mathbf{R}_y(\theta) \mathbf{R}_x(\phi)$$



Figure 4: YPR rotation sequence applied to an aircraft analogy.

1.6 Homogeneous Transformation Matrix (HTM)

HTM: Combined Rotation & Translation

The 4×4 Homogeneous Transformation Matrix encodes both **rotation** and **translation**:

$${}^A T_B = \begin{pmatrix} \underbrace{\mathbf{R}_{3 \times 3}}_{\text{Rotation}} & \underbrace{\vec{p}_{3 \times 1}}_{\text{Translation}} \\ \underbrace{\mathbf{0}_{1 \times 3}}_{0 \text{ row}} & \underbrace{1}_{1} \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Homogeneous coordinates: A 3D point $\vec{p} = (x, y, z)^T$ becomes $(x, y, z, 1)^T$ in homogeneous form.

1.6.1 Properties of HTM

Key Properties

1. **Composition:** ${}^A T_C = {}^A T_B \cdot {}^B T_C$

2. **Inverse:**

$${}^A T_B^{-1} = {}^B T_A = \begin{pmatrix} \mathbf{R}^T & -\mathbf{R}^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

3. **NOT commutative:** ${}^A T_B {}^B T_C \neq {}^B T_C {}^A T_B$ in general.

4. **Point transformation:** ${}^A \vec{p} = {}^A T_B \cdot {}^B \vec{p}$

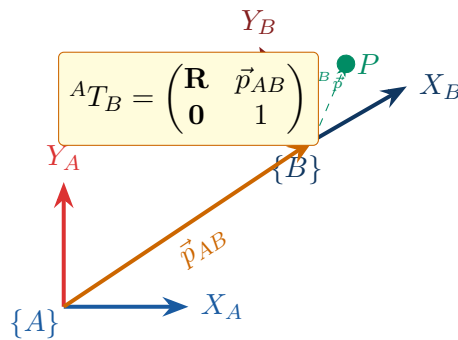


Figure 5: HTM ${}^A T_B$ encodes the position $\vec{p}_{AB}$ of frame $\{B\}$'s origin in $\{A\}$, plus the rotation $\mathbf{R}$.

Unit 2: Forward Kinematics

2.1 Link and Joint Parameters

Robot Structure

A robot arm consists of **rigid links** connected by **joints**:

- **Revolute joint**: Rotates by angle θ (like a door hinge).
- **Prismatic joint**: Translates by distance d (like a piston).
- **Link**: Rigid body connecting two consecutive joints.

An n -DOF robot has n joints and $n + 1$ links (including the base).

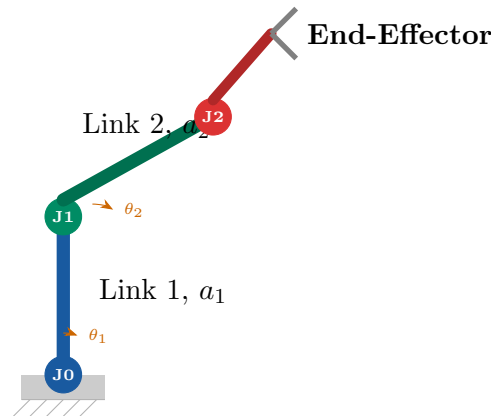


Figure 6: A 2-DOF planar robot with revolute joints J_0 and J_1 , links of length a_1 and a_2 .

2.2 Denavit–Hartenberg (DH) Parameters

The 4 DH Parameters

Each link-joint pair is described by **four parameters**:

Parameter	Description	Variable?
θ_i	Joint angle about Z_{i-1}	Yes (revolute joint)
d_i	Link offset along Z_{i-1}	Yes (prismatic joint)
a_i	Link length along X_i	No (fixed geometry)
α_i	Link twist about X_i	No (fixed geometry)

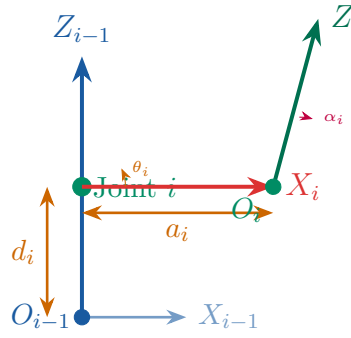


Figure 7: DH Parameters for a single link-joint pair. The four parameters define the transformation from frame $i - 1$ to frame i .

2.3 DH Transformation Matrix

The transformation from frame $\{i - 1\}$ to frame $\{i\}$ is:

$${}^{i-1}T_i = \underbrace{R_z(\theta_i)}_{\text{rotate about } Z_{i-1}} \cdot \underbrace{T_z(d_i)}_{\text{translate along } Z_{i-1}} \cdot \underbrace{T_x(a_i)}_{\text{translate along } X_i} \cdot \underbrace{R_x(\alpha_i)}_{\text{rotate about } X_i}$$

$${}^{i-1}T_i = \begin{pmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2.4 Forward Kinematics: Arm Matrix

Forward Kinematics (FK)

Given joint angles $\{\theta_1, \theta_2, \dots, \theta_n\}$, find the end-effector pose. The total transformation (Arm Matrix) is:

$${}^0T_n = {}^0T_1 \cdot {}^1T_2 \cdot \dots \cdot {}^{n-1}T_n$$

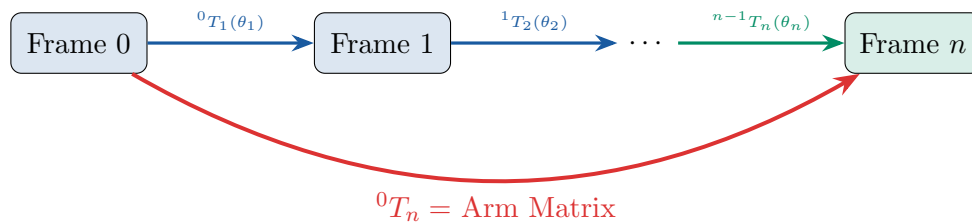


Figure 8: Chaining individual link transforms to obtain the overall FK arm matrix.

2.5 Workspace vs Joint Space

Two Important Spaces

- **Joint Space** (C-Space): The space of all joint variables $\vec{q} = (\theta_1, \theta_2, \dots, \theta_n)$. An n -dimensional space.
- **Task/Cartesian Space**: 3D physical space where the end-effector moves. Position (x, y, z) and orientation.
- **Workspace**: The set of all points reachable by the end-effector.
 - *Reachable workspace*: All points reachable in at least one orientation.
 - *Dexterous workspace*: Points reachable in **all** orientations.

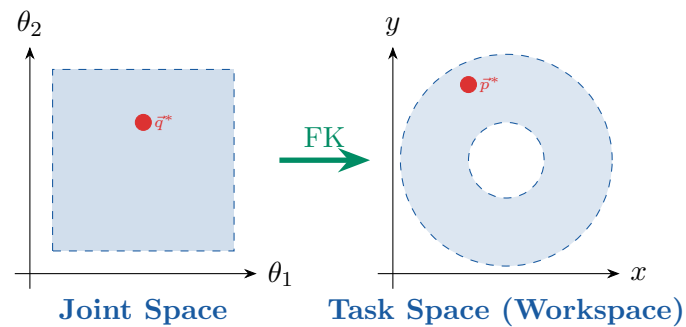


Figure 9: FK maps a point in joint space to a point in Cartesian task space. The workspace (annular region) shows all reachable positions for a 2-DOF robot.

Unit 3: Inverse Kinematics

3.1 Problem Definition

Inverse Kinematics (IK) Problem

Given: Desired end-effector position/orientation $\vec{x}_d = (x, y, z, \phi, \theta, \psi)^T$

Find: Joint angles $\vec{q} = (\theta_1, \theta_2, \dots, \theta_n)^T$ such that $\text{FK}(\vec{q}) = \vec{x}_d$

$$\vec{q} = \text{IK}(\vec{x}_d)$$

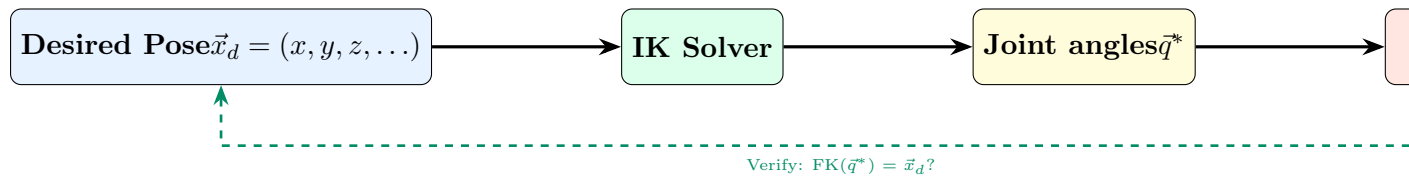


Figure 10: The IK problem: find joint angles that achieve a desired Cartesian pose. FK is the inverse map.

3.2 Solvability and Existence of Solutions

Solvability Conditions

1. **Existence:** A solution exists only if the target point is inside the robot's workspace.
2. **Uniqueness:** Multiple solutions often exist (e.g., elbow-up vs elbow-down).
3. **Infinite solutions:** When $\text{DOF} > \text{required DOF}$ (redundant robot).
4. **No solution:** Target outside workspace.

For a non-redundant robot: n DOFs required to specify n task variables.

3.3 Multiple Solutions: Elbow-Up vs Elbow-Down

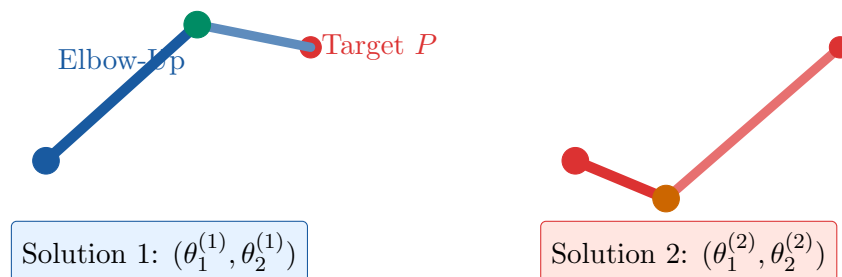


Figure 11: Two solutions for the 2-DOF IK problem reaching the same target P : elbow-up (left) and elbow-down (right).

3.4 Closed-Form vs Numerical Methods

Solution Methods

Closed-Form (Analytical) Methods:

- Derive explicit formulae: $\theta_i = f(\vec{x}_d)$.
- Fast, all solutions found, preferred for real-time control.
- Only possible for robots with special geometry (e.g., 6-DOF with wrist).
- Example: 2-DOF planar: $\theta_2 = \pm \arccos\left(\frac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}\right)$

Numerical (Iterative) Methods:

- Use the Jacobian to iteratively update $\vec{q}$ until error $\rightarrow 0$.
- Works for any robot, but may converge to local minimum.
- Slower; used when analytical solution unavailable.
- Examples: Newton-Raphson, Jacobian pseudoinverse method.

Unit 4: Differential Kinematics & Jacobian

4.1 Velocity Relationship

Joint Space $\leftrightarrow$ Task Space Velocity

Forward kinematics gives position: $\vec{x} = f(\vec{q})$.

Differentiating with respect to time:

$$\dot{\vec{x}} = \frac{\partial f}{\partial \vec{q}} \dot{\vec{q}} = \mathbf{J}(\vec{q}) \dot{\vec{q}}$$

where $\mathbf{J}$ is the **Jacobian matrix**, a $6 \times n$ (or $m \times n$) matrix.

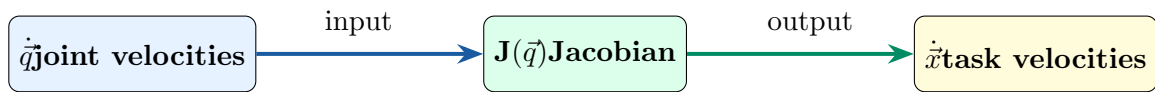


Figure 12: The Jacobian linearly maps joint velocities to end-effector velocities.

4.2 Jacobian Matrix

For a robot with n joints and m -dimensional task space:

$$\mathbf{J}(\vec{q}) = \begin{pmatrix} \frac{\partial x_1}{\partial q_1} & \cdots & \frac{\partial x_1}{\partial q_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_m}{\partial q_1} & \cdots & \frac{\partial x_m}{\partial q_n} \end{pmatrix}_{m \times n}$$

Example: 2-DOF Planar Robot Jacobian

For a 2-DOF robot with links a_1, a_2 :

$$x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2), \quad y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$$

$$\mathbf{J} = \begin{pmatrix} -a_1 \sin \theta_1 - a_2 \sin(\theta_1 + \theta_2) & -a_2 \sin(\theta_1 + \theta_2) \\ a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) & a_2 \cos(\theta_1 + \theta_2) \end{pmatrix}$$

4.3 Singularities

Singularities: When $\det(\mathbf{J}) = 0$

At a **singular configuration**, the Jacobian loses rank:

- The robot cannot move the end-effector in certain directions.
- Velocities tend to infinity near singularities.
- Should be avoided during trajectory planning.

Types:

- **Boundary singularity:** Arm fully extended or folded — at the boundary of workspace.
- **Interior singularity:** Two joint axes align — inside the workspace.

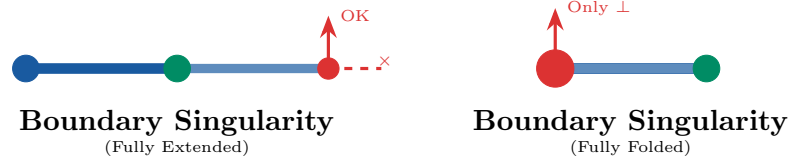


Figure 13: Singular configurations for a 2-DOF robot. At boundaries, certain motions become impossible.

Unit 5: Force Analysis & Statics

5.1 Force-Torque Relationship

Statics: Force and Torque Mapping

Just as the Jacobian maps joint velocities to end-effector velocities, the **transpose of the Jacobian** maps end-effector forces to joint torques, via the **principle of virtual work**.

$$\vec{\tau} = \mathbf{J}^T(\vec{q}) \vec{F}$$

where:

- $\vec{\tau} = (\tau_1, \tau_2, \dots, \tau_n)^T$ — vector of joint torques/forces
- $\mathbf{J}^T$ — transpose of the Jacobian
- $\vec{F} = (F_x, F_y, F_z, M_x, M_y, M_z)^T$ — end-effector wrench (force + moment)

5.2 Derivation via Virtual Work

Principle of Virtual Work

Virtual work done by joint torques equals virtual work done by end-effector forces:

$$\delta W = \vec{\tau}^T \delta \vec{q} = \vec{F}^T \delta \vec{x}$$

Since $\delta \vec{x} = \mathbf{J} \delta \vec{q}$:

$$\vec{\tau}^T \delta \vec{q} = \vec{F}^T \mathbf{J} \delta \vec{q} \Rightarrow \vec{\tau} = \mathbf{J}^T \vec{F}$$

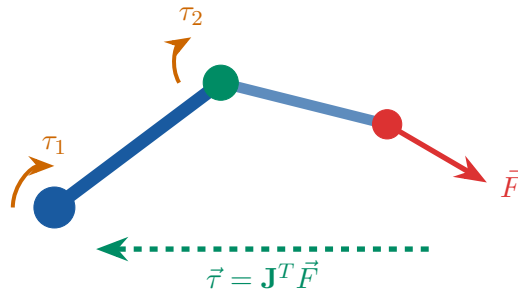


Figure 14: End-effector force $\vec{F}$ maps to joint torques τ_1, τ_2 via $\mathbf{J}^T$.

5.3 Key Insight: Jacobian Duality

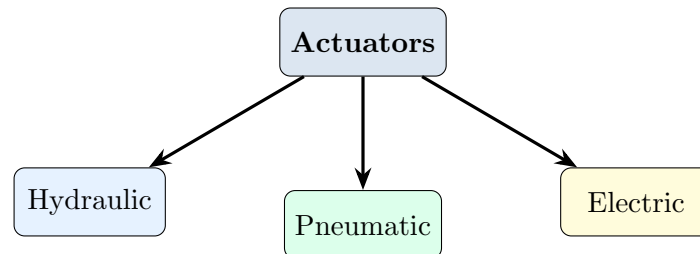
Duality of Jacobian

Relationship	Velocity	Force/Torque
Direction	Joint $\rightarrow$ Task	Task $\rightarrow$ Joint
Matrix	$\mathbf{J}$	$\mathbf{J}^T$
Equation	$\dot{\vec{x}} = \mathbf{J}\dot{\vec{q}}$	$\vec{\tau} = \mathbf{J}^T \vec{F}$

Near singularities: small end-effector forces require very large joint torques!

Unit 6: Actuators & Motors

6.1 Types of Actuators



Feature	Hydraulic	Pneumatic	Electric (DC)
Working fluid	Oil (incompressible)	Air (compressible)	Electricity
Power density	Very high	Medium	Medium-High
Precision	High	Low-Medium	Very High
Speed	Medium	Very fast	Fast
Cleanliness	Leaks possible	Clean	Clean
Typical use	Heavy robots, construction	Pick-and-place, grippers	Most industrial robots
Control	Hydraulic valves	Pneumatic valves	PWM / H-bridge

6.2 DC Electric Motors

DC Motor Principle

A DC motor converts electrical energy into mechanical rotation via the **Lorentz force**: a current-carrying conductor in a magnetic field experiences a force $\vec{F} = I(\vec{L} \times \vec{B})$.

- **Stator**: Fixed magnets providing $\vec{B}$ field.
- **Rotor (Armature)**: Rotating coil carrying current.
- **Commutator**: Reverses current direction to maintain continuous rotation.

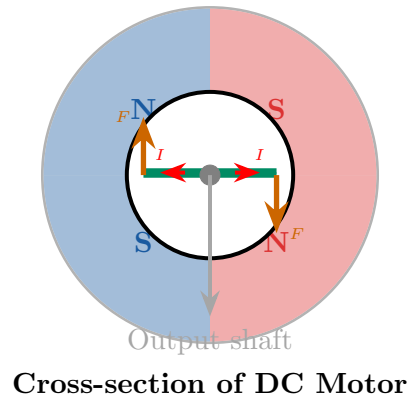


Figure 15: Cross-sectional view of a DC motor. Opposing forces on the armature conductors create a net torque (moment) rotating the shaft.

6.3 Torque-Speed Characteristics

The fundamental DC motor equations:

$$V = IR + k_e\omega \quad \tau = k_t I \quad \tau = \frac{k_t(V - k_e\omega)}{R}$$

where k_e = back-EMF constant, k_t = torque constant, ω = angular speed, R = armature resistance.

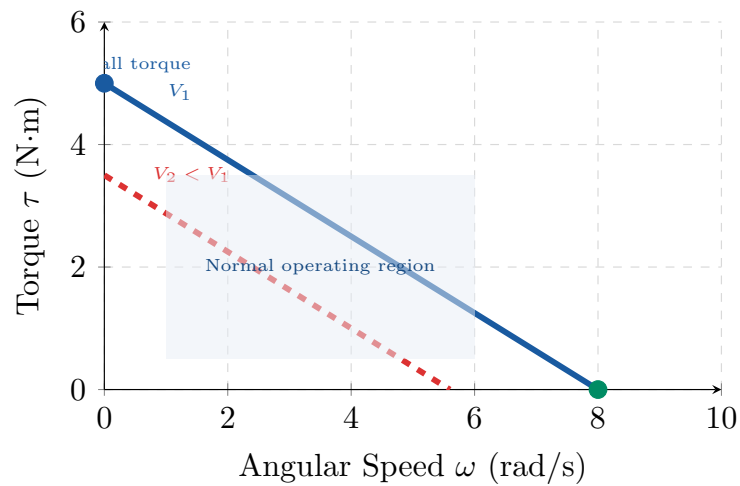


Figure 16: DC motor torque-speed curve. Torque is maximum at stall ($\omega = 0$) and zero at no-load speed ω_0 . Higher voltage shifts the curve up.

6.4 Motor Operational Considerations

Key Motor Selection Criteria

1. **Stall torque:** Maximum torque at zero speed. Critical for holding loads.
2. **No-load speed:** Maximum speed with no load. Determines top speed.
3. **Continuous torque rating:** Torque sustainable without overheating.
4. **Torque ripple:** Fluctuation in output torque due to commutation. Causes vibrations.
5. **Back-EMF:** Voltage generated opposing the supply; limits max speed.
6. **Efficiency:** $\eta = P_{\text{out}}/P_{\text{in}} = \tau\omega/(VI)$.
7. **Thermal limits:** Continuous operation raises winding temperature.

Unit 7: Circuit Basics & Control Architecture

7.1 Basic Circuit Elements

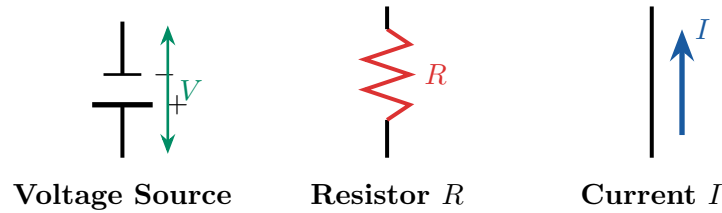


Figure 17: Basic circuit symbols: voltage source, resistor, and current direction.

7.2 Ohm's Law

$$V = I \cdot R \quad \Leftrightarrow \quad I = \frac{V}{R} \quad \Leftrightarrow \quad R = \frac{V}{I}$$

where V is in Volts, I in Amperes, R in Ohms (Ω).

7.3 Kirchhoff's Laws

KCL & KVL

Kirchhoff's Current Law (KCL): The algebraic sum of currents at any node is zero:

$$\sum_{\text{node}} I_k = 0 \quad \Leftrightarrow \quad \text{Current in} = \text{Current out}$$

Kirchhoff's Voltage Law (KVL): The algebraic sum of voltages around any closed loop is zero:

$$\sum_{\text{loop}} V_k = 0 \quad \Leftrightarrow \quad \text{Sum of rises} = \text{Sum of drops}$$



Figure 18: KCL at a node (left) and KVL around a loop (right).

7.4 Power in Circuits

$$P = V \cdot I = I^2 R = \frac{V^2}{R} \quad (\text{in Watts})$$

7.5 Switching Devices: MOSFET and IGBT

Power Switches for Motor Control

MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor):

- Voltage-controlled switch: Gate voltage V_{GS} controls channel (drain-source) current.
- Fast switching, low gate drive power.
- Best for low-to-medium voltage/current, high frequency (e.g., PWM up to 100s of kHz).

IGBT (Insulated Gate Bipolar Transistor):

- Combines MOSFET input characteristics with BJT output characteristics.
- Higher voltage/current capability than MOSFET.
- Slower than MOSFET but handles large power levels.
- Used in industrial drives, inverters.

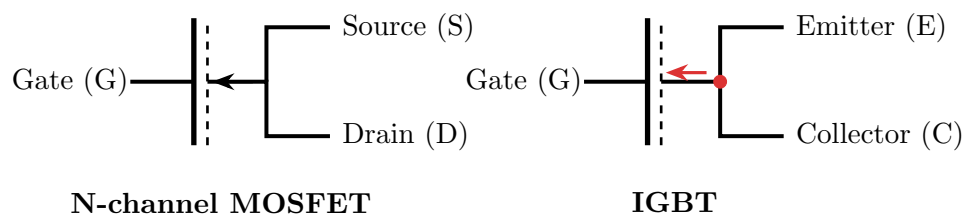


Figure 19: Simplified symbols for N-channel MOSFET and IGBT. Both are voltage-controlled switches used in motor drive circuits.

7.6 H-Bridge for DC Motor Control

H-Bridge Concept

An H-bridge uses **4 switches** (MOSFETs or IGBTs) arranged in an “H” shape to control the direction of current through a DC motor, enabling **forward**, **reverse**, and **braking**.

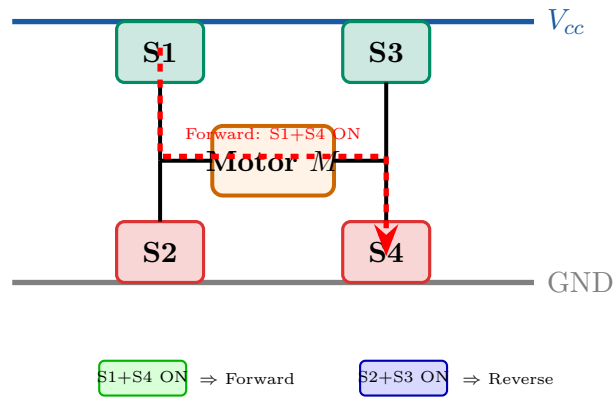


Figure 20: H-Bridge circuit. S1+S4 closed: current flows left-to-right through motor (forward). S2+S3 closed: current reverses (reverse). S1+S3 or S2+S4: braking.

7.7 PWM for Speed Control

Pulse Width Modulation (PWM)

By rapidly switching the H-bridge at a duty cycle D , the effective voltage seen by the motor is:

$$V_{\text{eff}} = D \times V_{cc}, \quad D = \frac{t_{\text{ON}}}{T}$$

$D = 100\% \rightarrow$ full speed; $D = 50\% \rightarrow$ half speed; $D = 0\% \rightarrow$ stop.

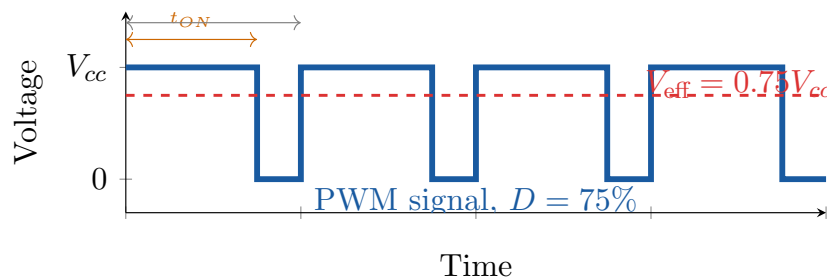


Figure 21: PWM signal with 75% duty cycle. The average voltage (dashed) determines effective motor speed.

Quick Reference Summary

Complete Formula Sheet

Unit 1 — Geometry & Transforms

$$\mathbf{R}_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{pmatrix}, \quad \mathbf{R}_y(\theta) = \begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix}, \quad \mathbf{R}_z(\psi) = \begin{pmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$T = \begin{pmatrix} \mathbf{R} & \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}, \quad T^{-1} = \begin{pmatrix} \mathbf{R}^T & -\mathbf{R}^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

Unit 2 — Forward Kinematics (DH)

$${}^{i-1}T_i(\theta_i, d_i, a_i, \alpha_i), \quad {}^0T_n = \prod_{i=1}^n {}^{i-1}T_i$$

Unit 3 — Inverse Kinematics

$$\vec{q} = \text{IK}(\vec{x}_d), \quad \theta_2 = \pm \arccos\left(\frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}\right)$$

Unit 4 — Jacobian

$$\dot{\vec{x}} = \mathbf{J}(\vec{q})\dot{\vec{q}}, \quad \text{Singular when } \det(\mathbf{J}) = 0$$

Unit 5 — Force Statics

$$\vec{\tau} = \mathbf{J}^T \vec{F}$$

Unit 6 — DC Motor

$$V = IR + k_e \omega, \quad \tau = k_t I, \quad \tau = \frac{k_t(V - k_e \omega)}{R}, \quad P = \tau \omega$$

Unit 7 — Circuits

$$V = IR, \quad P = VI = I^2 R, \quad \sum I = 0 \text{ (KCL)}, \quad \sum V = 0 \text{ (KVL)}, \quad V_{\text{eff}} = DV_{cc}$$

Exam Tips

- **HTM inverse:** Remember $\mathbf{R}^{-1} = \mathbf{R}^T$ and how the translation transforms.
- **DH convention:** Assign frames systematically — Z_i along joint $i + 1$ axis, X_i along common normal.
- **Multiple IK solutions:** Always state elbow-up and elbow-down; check workspace.
- **Jacobian:** Derive by differentiating FK w.r.t. each joint angle.

- $J^T F$: Virtual work derivation is expected — know the two-line proof.
- **H-bridge**: Know which switches to close for forward, reverse, and brake.
- **Both units**: Equal weightage — don't neglect electrical topics!